

CoRoT-2b

R-1566

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_150618 · created 2026-08-01T22:03:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457191.9168 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.069 +/- 0.053
Transit depth (Rp/Rs)^2	0.48 +/- 0.73 [%]
Orbital Inclination (inc)	82.94 +/- 2.11
Airmass coefficient 1 (a1)	1.093 +/- 0.094
Airmass coefficient 2 (a2)	-0.09 +/- 0.21
Scatter in the residuals of the lightcurve fit is	8.56 %
Best Comparison Star	#3 - [468, 403]
Optimal Aperture	2.92
Optimal Annulus	10.66
Transit Duration (day)	0.049 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.069 ± 0.053 vs 0.1667 (deviation 1.84σ)
Duration fit vs published	0.049 d vs 0.095 d (deviation 1.64σ)
Mid-time O–C	0.6 min
Depth SNR	1.3
Residual scatter	8.56 %

Light curve

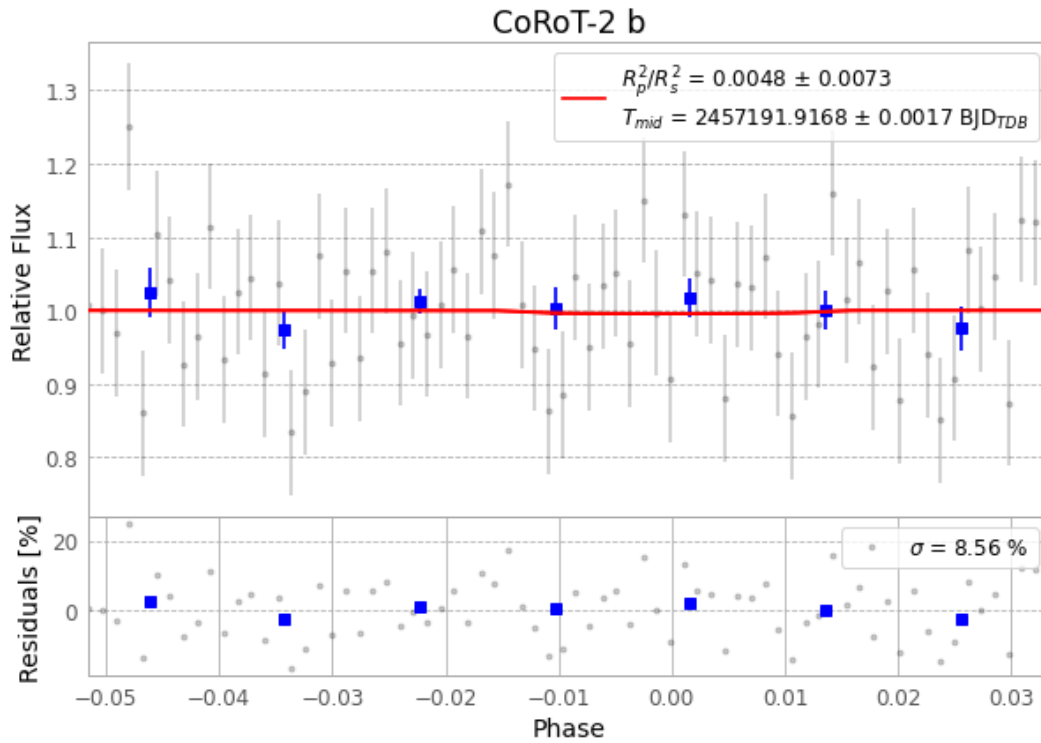


Figure 1 — FinalLightCurve_CoRoT-2 b_2015-06-18.png (SHA-256 72a0bf726ca23477...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [330, 186], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bf10f39a2e244db9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), CoRoT-2150618074212.FITS → CoRoT-2150618111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-150618.log (4772f59ac9f3...), FinalLightCurve_CoRoT-2 b_2015-06-18.png (72a0bf726ca2...), FinalLightCurve_CoRoT-2 b_2015-06-18.csv (99bf58f7f042...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)